Skills Summary

Finding and Verifying Inverse Functions

Use this reference while completing your worksheet or when studying.

What an inverse is. f^{-1} undoes f: if $f(a) = b$ then $f^{-1}(b) = a$. So $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$.

Notation warning. f^{-1} is the inverse *function*. It is not $\dfrac{1}{f(x)}$.

Domain and range swap. The domain of f^{-1} is the range of f, and its range is the domain of f. Graphically, f^{-1} is the reflection of f across the line $y = x$.

Only one-to-one functions have inverses. If a horizontal line meets the graph of f twice, two inputs share an output and there is nothing for f^{-1} to return — restrict the domain first.

1. Find the inverse: swap, then solve

Write $y = f(x)$; swap x and y; solve the new equation for y; rename it $f^{-1}(x)$. Solving means undoing the operations **in reverse order** — whatever was done last to x is undone first.

Example: Find the inverse of $f(x) = 4x + 3$.
Step 1. The question asks which input produces a given output, so swap the roles of the variables and solve for the new y.

Step 2. $y = 4x + 3$ becomes $x = 4y + 3$. Subtract 3: $x - 3 = 4y$. Divide by 4: $y = \dfrac{x-3}{4}$.

$f^{-1}(x) = \boldsymbol{\dfrac{x-3}{4}}$

Try it: Find the inverse of $f(x) = 6x - 1$.

check: $\boldsymbol{f^{-1}(x) = \frac{x+1}{6}}$

2. Verify a pair with composition

Two functions are inverses only when **both** $f(g(x)) = x$ and $g(f(x)) = x$. Substitute one whole function into the other and simplify; the variable must survive alone, with every constant cancelling.

Example: Are $f(x) = \dfrac{x}{2} + 4$ and $g(x) = 2x - 8$ inverses?
Step 1. "Verify" means composition, not re-deriving — substitute g into f and simplify, then repeat the other way round.

Step 2. $f(g(x)) = \dfrac{2x-8}{2} + 4 = (x - 4) + 4$, and the constants cancel. The composition collapses to $\boldsymbol{x}$, and $g(f(x))$ does the same, so the pair is inverse.

Try it: Simplify $f(g(x))$ for $f(x) = 3x - 1$ and $g(x) = \dfrac{x+1}{3}$. What does it come out to?

check: $\boldsymbol{x}$ — so they are inverses

3. Inverse values, domain and restriction

$f^{-1}(b)$ asks: which input of f gives the output b? You can answer it either by building the formula for f^{-1} and evaluating, or by solving $f(x) = b$ directly — the two routes must agree. When f is not one-to-one, restrict its domain first (for $f(x) = x^2$ take $x \geq 0$); the inverse then inherits its domain from the range of the restricted f.

Example: For $f(x) = 2x + 7$, find $f^{-1}(9)$.
Step 1. Only one value is wanted, so either build the inverse formula or solve $f(x) = 9$ — the formula route also leaves you something reusable.
Step 2. $x = 2y + 7$ gives $f^{-1}(x) = \dfrac{x-7}{2}$, so $f^{-1}(9) = \dfrac{9-7}{2} = \dfrac{2}{2}$. $f^{-1}(9) = \mathbf{1}$, and indeed $f(1) = 2 + 7 = 9$.

Try it: For $f(x) = 5x - 2$, find $f^{-1}(13)$. Check your answer by putting it back into f.

check: **3**, since $f(3) = 13$

(!) Watch out: undo operations in reverse order. For $f(x) = 3x + 5$ the inverse is $\dfrac{x-5}{3}$, not $\dfrac{x}{3} - 5$ — the 5 comes off before the division, because it was added last.